

BUSINESS_RULES.md

Version: 1.0

Status: Mandatory

Classification: Business Logic Constitution

Applies To:

- AI Engine
 - Order Processing
 - WhatsApp Integration
 - WooCommerce Integration
 - Dashboard
 - Notifications
 - Admin Portal
 - Customer Portal
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1. Purpose

This document defines the official business rules for CommercePilot.

All business logic must follow these rules.

If implementation conflicts with this document, this document takes precedence.

2. Business Philosophy

CommercePilot automates repetitive operational work.

CommercePilot never replaces business decision-making.

Artificial Intelligence assists.

Humans decide.

3. Human-in-the-Loop Principle

AI may:

- Read messages

- Understand customer intent
- Extract order information
- Match products
- Calculate confidence
- Recommend actions
- Draft replies

AI may NOT:

- Approve high-risk orders
- Change inventory manually
- Delete records
- Modify business settings
- Override owner decisions

The business owner always has final authority.

4. Tenant Rules

Every business is a separate tenant.

Each tenant has isolated:

- Users
- Customers
- Products
- Orders
- Messages
- Analytics
- AI Context
- Settings

Cross-tenant access is strictly prohibited.

5. Customer Rules

A customer is uniquely identified by their WhatsApp phone number within a tenant.

If a new message is received:

- Existing customer → attach conversation
- New customer → create customer profile automatically

Customer history must never be lost.

6. WhatsApp Message Rules

Every incoming message must be stored before processing.

Messages are immutable.

Supported types:

- Text
- Image (Future)
- Voice (Future)
- Document (Future)

Each message receives:

- Message ID
 - Tenant ID
 - Customer ID
 - Timestamp
 - Processing Status
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7. AI Processing Rules

Every message enters the AI pipeline.

Processing steps:

1. Validate webhook
2. Save message
3. Detect intent
4. Retrieve product catalog (RAG)
5. Extract entities
6. Calculate confidence
7. Create draft result
8. Notify owner (if required)

AI must never skip validation.

8. Intent Detection Rules

Supported intents:

- Order
- Product Inquiry
- Price Inquiry
- Stock Inquiry

- Greeting
- Delivery Inquiry
- Complaint
- Support
- Unknown

Unknown intents require manual review.

9. Product Matching Rules

AI must only match products from the tenant's catalog.

Never invent products.

If multiple products match:

Return the best matches with confidence.

If no product matches:

Request clarification from the customer.

10. Confidence Score Rules

Confidence Score Range:

0.00 – 1.00

0.95–1.00 Automatic draft creation.

0.80–0.94 Owner review recommended.

Below 0.80 Manual confirmation required before order creation.

Confidence score must be stored with every AI decision.

11. Order Creation Rules

Orders are created as:

Draft

↓

Pending Owner Approval

↓

Approved

↓

Synced to WooCommerce

↓

Customer Confirmed

↓

Completed

AI never creates a final order directly.

12. Order State Machine

Allowed transitions:

Draft

↓

Pending Approval

↓

Approved

↓

Processing

↓

Completed

Alternative paths:

Pending Approval

↓

Rejected

Approved

↓

Cancelled

Completed orders cannot return to earlier states.

13. Owner Approval Rules

The owner may:

- Approve
- Reject
- Edit
- Request clarification

Every owner action must be logged.

Owner approval overrides AI recommendations.

14. Inventory Rules

Before approval:

Validate stock.

If stock is unavailable:

Flag the order.

Notify the owner.

Suggest alternatives (future feature).

Inventory is never reduced until the order is officially approved and synchronized.

15. WooCommerce Synchronization Rules

Synchronization occurs only after owner approval.

If synchronization fails:

Retry automatically.

Log the failure.

Notify the owner after retry limits are exceeded.

Never create duplicate WooCommerce orders.

Idempotency must be enforced.

16. Notification Rules

Notify the owner when:

- New draft order created
- AI confidence is low
- WooCommerce sync fails
- AI processing fails

Notify the customer when:

- Order received
 - Order approved
 - Order rejected
 - Additional information required
 - Order shipped (future)
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17. AI Reply Rules

AI replies must be:

Professional

Friendly

Concise

Context-aware

AI must never:

Promise unavailable products.

Guess delivery dates.

Guess prices.

Invent discounts.

18. Duplicate Order Prevention

Duplicate detection considers:

- Customer
- Product
- Quantity
- Recent conversation
- Time window

Potential duplicates require owner review.

19. Audit Rules

Every business event is recorded.

Audit entries include:

- Timestamp
- Tenant ID
- User ID
- Customer ID
- Order ID
- Event Type
- Previous State
- New State

Audit logs are immutable.

20. Automation Rules

Automation is allowed only for low-risk repetitive tasks.

Examples:

✓ Message storage

✓ AI extraction

✓ Email sending

✓ Notifications

✓ Draft order generation

Automation is prohibited for:

✗ Final approval

✗ Deleting records

✗ Price overrides

✗ Refunds

✗ Business settings

21. Error Handling Rules

If AI fails:

Save the message.

Notify the owner.

Allow manual processing.

If WooCommerce fails:

Retry using queue.

Do not lose data.

Never silently ignore failures.

22. Retry Policy

Retry:

AI API failures

Webhook delivery failures

WooCommerce failures

Email failures

Maximum retries:

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Use exponential backoff.

After maximum retries:

Create a manual intervention task.

23. Business Metrics

Track:

- Orders processed
 - AI accuracy
 - Average approval time
 - WooCommerce sync success
 - AI confidence distribution
 - Customer response time
 - Manual intervention rate
 - Automation success rate
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24. AI Learning Rules

CommercePilot does not automatically retrain AI using customer conversations.

Business data belongs to the tenant.

No tenant data may be shared with another tenant.

Future model improvements require explicit owner consent where applicable.

25. Business Rule Change Policy

Business rules are version-controlled.

Changes require:

- Documentation updates
- Testing
- Approval
- Changelog entry

26. Definition of Business Correctness

A feature is business-correct only if:

- ✓ It follows all business rules.
- ✓ It respects tenant isolation.
- ✓ It supports owner approval.
- ✓ It never invents products.
- ✓ It maintains audit logs.
- ✓ It preserves customer history.
- ✓ It prevents duplicate orders.
- ✓ It integrates safely with WooCommerce.
- ✓ It supports future platform expansion.

Final Principle

CommercePilot exists to automate repetitive business operations while preserving human control over critical business decisions.

Every workflow should reduce manual effort, improve accuracy, and maintain trust.

Whenever business efficiency and business safety conflict, business safety takes priority.